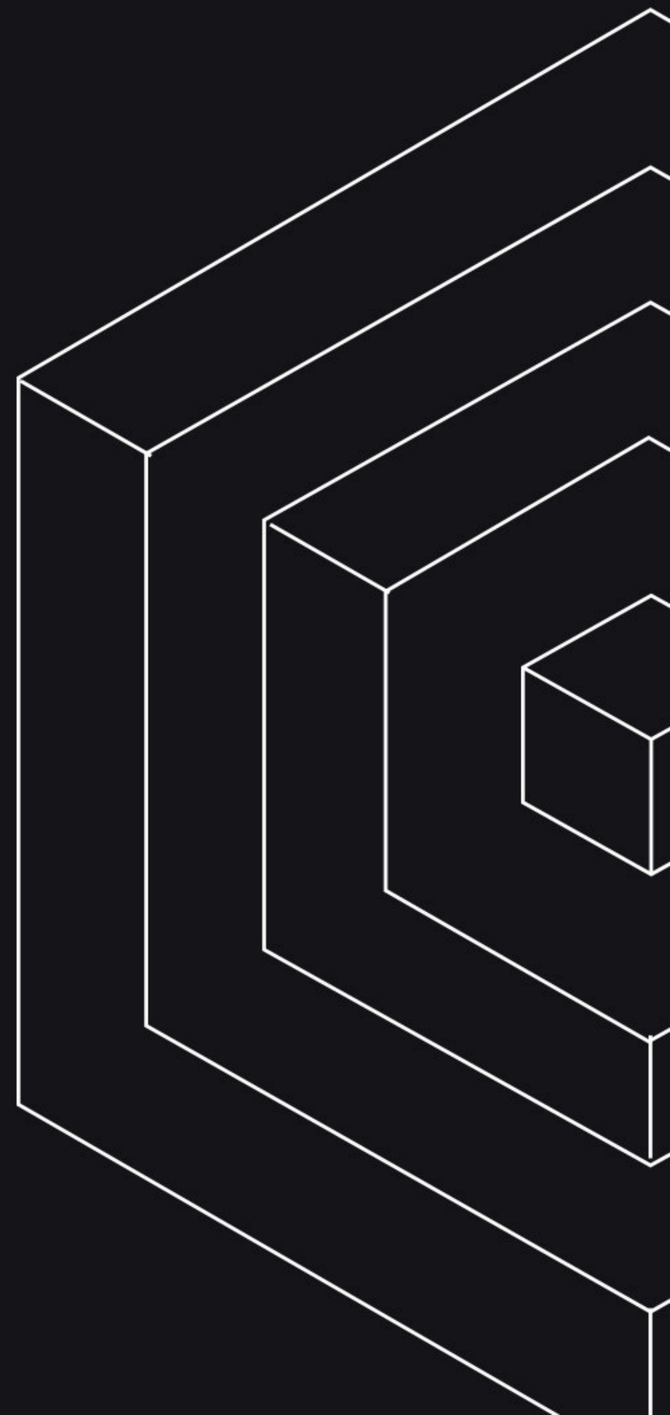
CORTEX LABS

Autonomous Intelligence Platform for Enterprise AI



End-to-End AI Ops Pipeline: Building a scalable intelligence engine for model deployment, monitoring, and continuous learning

Cortex Labs

The Enterprise AI Gap

Cortex Labs operates across the rapidly evolving enterprise AI landscape — where model deployment, observability, and governance determine which organizations lead and which fall behind.

- Model Sprawl: Teams ship models faster than they can monitor, creating blind spots across production
- Governance Gaps:
No unified audit trail from experiment to production deployment
- Fragmented Toolchains:
ML pipelines stitched together with duct tape across disconnected platforms
- Talent Bottleneck:
ML engineers spend 70% of time on infrastructure, not intelligence

Primary Objective

AI Operations streamlining model lifecycle management, ensuring production reliability, and building long-term institutional AI capability across the organization.



Platform Goals

Accelerate Deployment

Ship models to production in hours, not weeks

Ensure Reliability

Detect drift and degradation before they impact decisions

Scale Intelligence

Build a continuous, self-improving AI operations engine

OPTION 1

Managed MLOps Pipeline

A turnkey, cloud-native deployment system built on proven infrastructure — no custom orchestration required.

Model Registry

Versioned model storage and experiment tracking

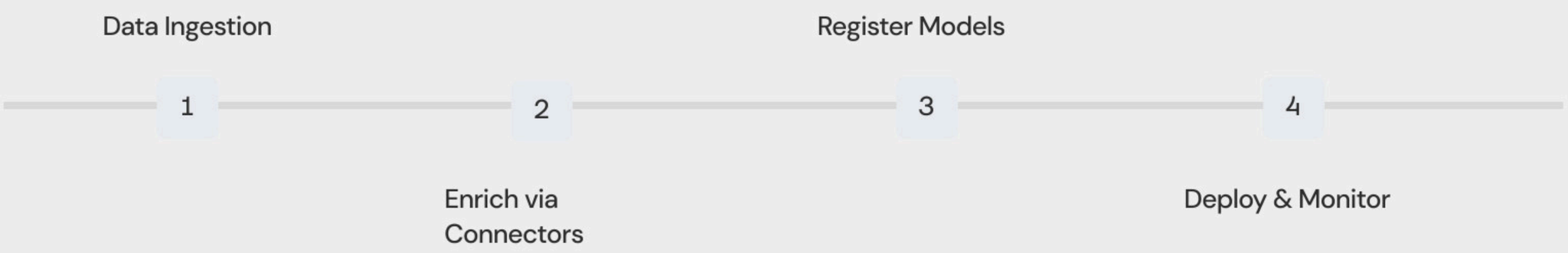
Inference Engine

Auto-scaling serving infrastructure with A/B routing

Data Connectors

Ingestion pipelines for streaming and batch data sources

Workflow



OPTION 1 ASSESSMENT

Managed MLOps: Strengths & Limitations

ADVANTAGES

Rapid Onboarding

Production-ready within days on existing cloud infrastructure

Automated Pipelines

Scalable training and inference workflows via managed services

Low Overhead

No new platforms — works within current DevOps workflows

LIMITATIONS

Limited Observability

Model performance and drift signals scattered across disconnected dashboards

Manual Tuning

Hyperparameter optimization and retraining remain time-intensive

Knowledge Silos

Experiment insights fragmented across notebooks and repos

Cortex Labs AI Operations Platform

Powered by Cortex Engine, a unified control plane designed to connect model registry, inference, monitoring, and governance into a single intelligent system.

- Model Registry: Versioning & tracking
- Cortex Engine:
  Operational intelligence hub
- Inference Engine:
  Real-time and batch serving
- Data Layer:
  Feature stores / Vector DBs / Streams

Model & Pipeline Intelligence:
Centralized dashboards for models, pipelines, and deployments across all environments with real-time health signals.

Strategic Value for Enterprise AI

Earlier detection of model degradation
Institutional knowledge preserved across teams

Core Platform Capabilities

OPTION 2 ARCHITECTURE

Platform Architecture

Cortex Engine becomes the central control plane for AI teams — connecting data sources, model registries, and deployment targets into a unified operational layer.

Why Centralization Matters

In competitive AI markets, speed and reliability win. A centralized architecture ensures every team member operates from the same intelligence — eliminating duplicate work and accelerating time to production.

Single Source of Truth

All model metadata and experiments live in one place

Real-Time Collaboration

Teams work from the same data simultaneously

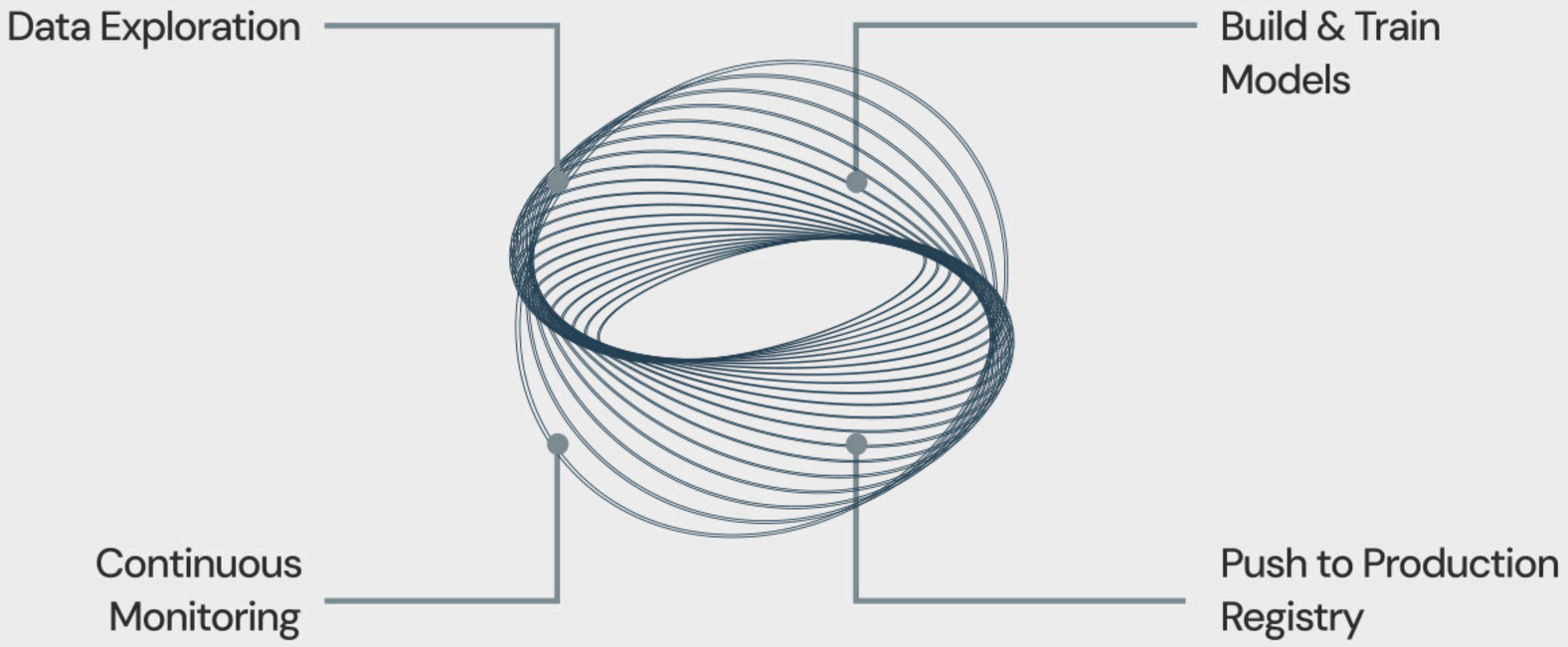
Seamless Handoffs

From research to training to production without gaps

OPTION 2 WORKFLOW

AI Ops Lifecycle with Cortex Engine

A continuous intelligence loop — from data exploration to production monitoring, fully connected.

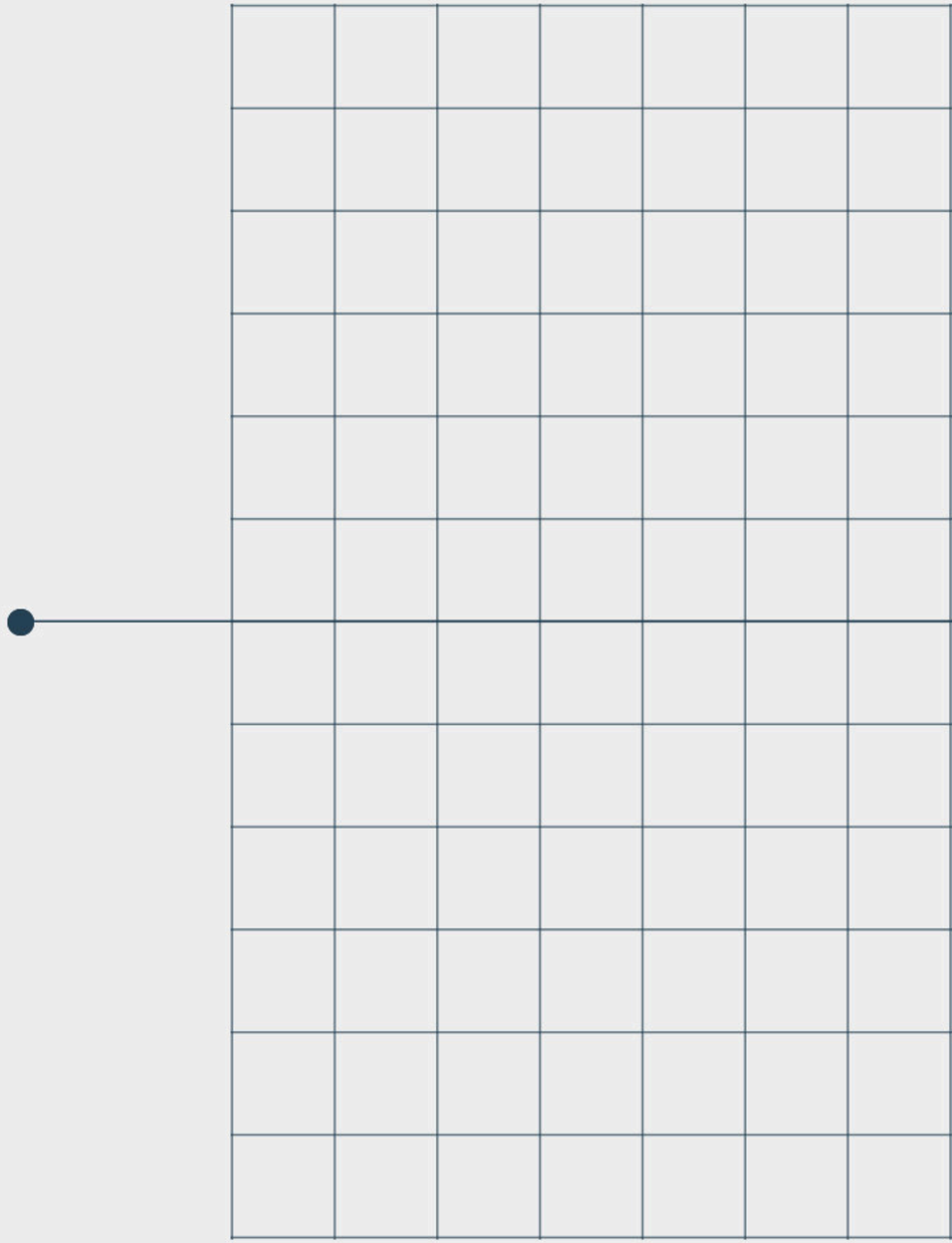


Unlike a linear process, this loop continuously feeds production signals back into training — ensuring models improve with every deployment cycle.

SECURITY, GOVERNANCE &
COMPLIANCE

Cortex Engine Capabilities

- Model Observatory: Centralized dashboards for every model in production, enriched with performance metrics, drift detection, and usage analytics.
- AI-Powered Insights: Automatically surface anomalies, suggest retraining triggers, and correlate model behavior with business outcomes.
- Deployment Workspace: Prepare canary rollouts, A/B experiments, and blue-green deployments with full context before pushing to production.
- Operations Dashboards: Track pipeline health, inference latency, and resource utilization in real time.



Model lineage and audit trails stay within the platform — no longer scattered across notebooks, repos, or individual experiment logs.

Option Comparison

Both options accelerate AI operations. The right choice depends on the team's appetite for depth and long-term scalability.

Feature	Managed MLOps	Cortex Engine Platform
Model Registry	✓	✓
Inference Serving	✓	✓
Data Enrichment	✓	✓
Model Research Workspace	✗	✓
Knowledge Management	Limited	Centralized
Drift Detection & Monitoring	✗	✓
Team Collaboration	Basic	Advanced

Expected Outcomes

Both options drive AI operations forward — Cortex Engine elevates it to an enterprise-wide competitive advantage.

MANAGED MLOPS

Fast-Track Results

1. Rapid Model Deployment: Launch models to production endpoints across target environments rapidly
2.
3. Standard Monitoring: Basic health checks and alerting on inference endpoints

CORTEX ENGINE

Strategic Infrastructure

1. Intelligent Operations Infrastructure: A repeatable, scalable system purpose-built for enterprise AI ops
2.

TIMELINE

Implementation Timeline

A Six-Week Sprint From Architecture To Production Intelligence — Structured To Minimize Disruption While Maximizing Operational Impact.

Week 1-2

Platform Architecture Setup And Cloud Infrastructure Configuration

Week 3-4

Model Registry Integration And Initial Pipeline Creation

Week 5

Inference Infrastructure Setup And Load Testing

Week 6

Production Launch, Monitoring Activation, And Iteration

NEXT STEPS

Pick one high-impact model. We ship the first pipeline.

Define Priority Models: Align on critical AI workloads, model architectures, and deployment targets for initial rollout.

